

Part B: Calculation Exercise:

Exercise 1: Count Vector:

$$D1: [1, 0, 1, 1, 0]$$

$$D2: [0, 1, 1, 1, 0]$$

$$D3: [1, 0, 0, 1, 1]$$

Exercise 2: TF:

$$a, \text{tf}(\text{cat}, D1) = \frac{1}{3}$$

$$b, \text{tf}(\text{eats}, D1) = \frac{1}{3}$$

$$c, \text{tf}(\text{fish}, D1) = \frac{1}{3}$$

Exercise 3: Corpus has $N=3$ and $\text{df}(\text{cat}) = 2$

$$\text{df}(\text{dog}) = 1$$

$$\text{df}(\text{eats}) = 2$$

$$\text{df}(\text{fish}) = 3$$

$$\text{df}(\text{likes}) = 1$$

$$\text{idf}(\text{cat}) = \log \frac{N}{\text{df}(\text{cat})} = \log \frac{3}{2} \approx 0,405$$

$$\text{idf}(\text{eats}) = \log \frac{3}{2} \approx 0,405$$

$$\text{idf}(\text{dog}) = \log \frac{3}{1} \approx 1,099$$

$$\text{idf}(\text{fish}) = \log \frac{3}{3} = 0$$

$$\text{idf}(\text{likes}) = \log \frac{3}{1} \approx 1,099$$

Term "fish" appears in all 3 documents

so $\text{df}(\text{fish}) = N = 3$ and $\text{idf}(\text{fish}) = 0$

$\Rightarrow$ Term "fish" cannot help distinguish between documents

Exercise 4: TF - IDF:

$$\text{tf} \cdot \text{idf}(\text{cat}, D1) = \text{tf}(\text{cat}, D1) \cdot \text{idf}(\text{cat}) = \frac{1}{3} \cdot \log \frac{3}{2} \approx 0,0586$$

$$\text{tf} \cdot \text{idf}(\text{eats}, D1) = \text{tf}(\text{eats}, D1) \cdot \text{idf}(\text{eats}) = \frac{1}{3} \cdot \log \frac{3}{2} \approx 0,0586$$

$$\text{tf. idf (fish, D1)} = \text{tf (fish, D1)} \cdot \text{idf (fish)} = \frac{1}{3} \cdot 0 = 0$$

Exercise 5: Cosine Similarity:

$$\begin{aligned} x &= [1, 1, 1] \\ y &= [0, 0, 0] \end{aligned} \Rightarrow \cos(x, y) = \frac{x^T y}{|x| \cdot |y|}$$

Tích vô hướng: $x^T \cdot y = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 = 2$

Độ dài x và y : $|x| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

$$|y| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$$

Vậy $\cos(x, y) = \frac{2}{\sqrt{6}} \approx 0,8165$

Part C: Prediction:

$D1 = \overset{\text{medical}}{\text{"image classification"}} ; D2 = \text{"medical image analysis"}$

$D3 = \text{"natural language processing"}$

Query = "medical image classification"

1. Document has the highest similarity: $D1$

2. Document has the lowest similarity: $D3$

3. Terms may have low IDF values: "medical" and "image"

4. If IDF is removed, only count vectors used cannot change the ranking

Prediction 1: If corpus contains 30K documents, the vocabulary may contain approximately 600,000 ^{unique} terms.

Prediction 2: The TF-IDF matrix will be dense or sparse? Sparse

How large can the percentage of zero entries be? > 70%

Prediction 3: For an arbitrary query, are the documents at the top of the search results necessarily the documents that are closest in meaning? Not necessarily